

ISABELLA FEHLE

(847) 313-8372 | ifehle21@gmail.com | Authorised to work in the USA, EU, and UK

PROFESSIONAL SUMMARY

I deliver data-driven solutions that address complex challenges across diverse projects. As an SME, I work with cross-functional teams to implement initiatives that align with organizational goals. I combine my technical expertise with the ability to translate complex data into clear business impact, effectively communicating risks, trade-offs, and projected outcomes. I have extensive experience designing and implementing MES configuration changes, as well as working with business management systems such as SAP and Windchill.

PROFESSIONAL EXPERIENCE

Abbott Laboratories, Neuromodulation (NMD)

Plano, TX

July 2025 - Present

Manufacturing Process Engineer

- Oversee \$1.2M+ in weekly production output, analyze scrap trends, and lead root cause analyses (RCAs) to improve efficiency and maintain product quality
- Primary engineering point-of-contact for production line, driving design tooling, maintain equipment uptime, and reconfigure line layouts to eliminate bottlenecks
- Lead cross-functional efforts to deploy and validate new equipment, materials, and processes as part of CAPA and new product development efforts
- Conduct testing to ensure processes are in control and compliant with regulations, collaborating cross-functionally with suppliers, quality, and design teams to resolve non-conformances
- Utilize statistical process control, FMEA, and other quality tools to monitor and improve product and process quality

Abbott Laboratories, Cardiac Rhythm Management (CRM)

Sylmar, CA

Jan 2024 - July 2024

Operations Co-Op

- Conducted 3 investigations across 2 different product lines to improve production and minimize nonconformances using lean six sigma methods
- Designed and implemented 3 new production tools using SolidWorks and Advanced Creo Parametric
- Validated installation, operation, and performance of production equipment
- Routed 50+ document changes using Windchill PLM system

Guilak Lab, Department of Orthopaedic Surgery, WashU in St. Louis School of Medicine

St. Louis, MO

May 2023 - Jan 2024

Osteoarthritis Research Assistant

- Developed and conducted experiments with focuses in cell culture, genetic circuit editing, and viral transduction
- Analyzed and evaluated experimental data using fluorescence and luminescence imaging
- Research Published: Kim, Y. S., Steward, N., Kim, A., **Fehle, I.**, & Guilak, F. (2024). Tuning the Response of Synthetic Mechanogenetic Gene Circuits Using Mutations in TRPV4. *Tissue engineering. Part A*

Washington University in St. Louis, McKelvey School of Engineering

St. Louis, MO

Aug 2022 - May 2025

Problem Solving Team Leader | Introduction to Biomedical Engineering Course

- Mentored a group of 10+ first-year biomedical engineering students
- Taught practice problems and introduced students to MATLAB, Python, and CAD Autodesk Inventor

Cardiovascular Translational Research Center, University of South Carolina

Columbia, SC

July 2022 - Sep 2022

Laboratory Assistant

- Calibrated and maintained instruments to keep equipment fully operational with a focus on pressure myography
- Cleaned and prepared supplies for experiments and measured compounds and solutions for use in testing

EDUCATION

Washington University in St. Louis

St. Louis, MO

Bachelor of Science in Biomedical Engineering & Minor in Mechanical Engineering

2021 - 2025

Honors: Dean's List | Overall GPA: 3.7/4.0

Senior Design Project: Designed a wearable system to monitor gait and predict early ALS symptoms using high frequency data, delivering a prototype device

Society of Women Engineers: Director of Professional Development, Director of Social Programming, Secretary, Coordinated with SWE board members and other groups on campus to host events for 250+ members

SKILLS

- Statistical Process Control, Root Cause Analysis, Validation and Verification, SolidWorks, Creo, Autodesk Inventor, MATLAB, Java, Python, R, Microsoft Office, Data Analysis, Cell Culture Techniques